BIOCHEMISTRY CUMULATIVE EXAM

January 21, 2000

1. (20 pts) Given the peptide, His-Asp-Arg-Tyr-Val-Pro-Glu-Phe-Cys. What is the charge of the peptide at pH 4 and at pH 10? Answers should be within 1/2 charge unit. TO RECEIVE CREDIT, SHOW WORK.

Use these pKa values:

terminal COOH 4
terminal NH_3^+ 9
R-GROUPS:
Asp 4
Glu 4
Cys 8
Tyr 10
His 6
Lys 10
Arg 12

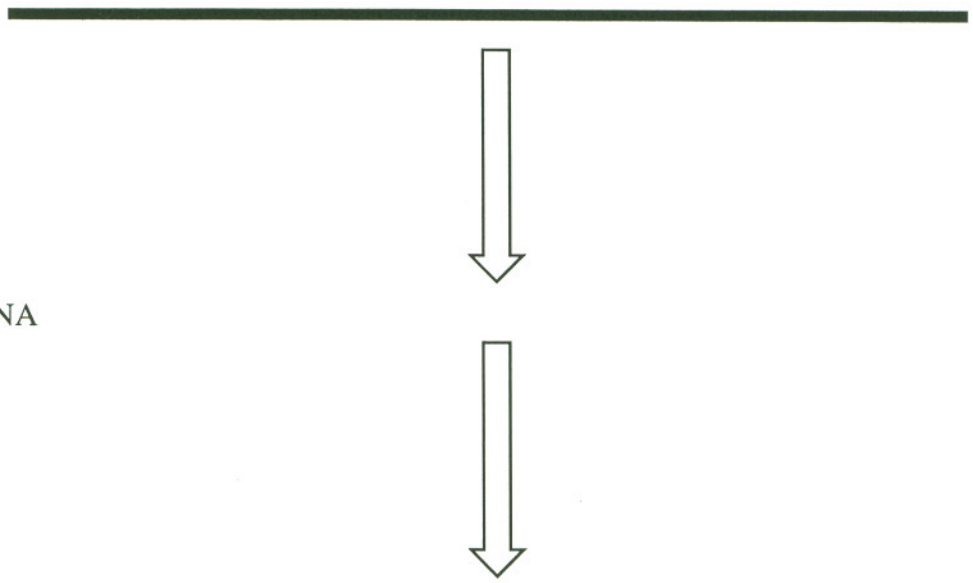
2 (30 pts) Name three zymogens. Briefly describe their function. Why is it necessary or advantageous for the body to make each one of these as zymogen?

3. (50pts) Below is a prototypical bacterial gene. Label the relative positions of:
Termination codons TGA TAA
–10 box
-35 box
AUG start codon
Repressor coding site
Transcriptional start site
Hairpin

Draw an RNA transcript of this gene in the space labeled RNA. Align the beginning and the end of this transcript with corresponding regions of the gene and label the 5' and 3' ends.

Also draw the polypeptide product of this gene in the space labeled protein. Align the beginning and the end of the polypeptide with the corresponding regions of the RNA transcript and indicate the amino and carboxy termini.

Gene



RNA

Protein